CSE499B – Section 15 – Group 03
Md. Rahad Hossain, Md. Yousuf, MD. Rokib Hasan Oli, Safayat Ibrahim

CityPulse: Intelligent Parking & Traffic Control Platform

Weekly Progress Report – Update 4 (Sprint 4: Plate Reads, Booking Lifecycle and Accounts)

I. Sprint Overview

This update covers 4 – 11 November 2026. The plate models now run inside the AI service, and the backend checks every parked car's plate against its booking. For the AI contribution, a method proposal for our own plate detection and reading models was written from an error study and 22 papers. A booking can now be priced before it is made, checked in, extended, checked out and paid, and old bookings are closed automatically. Users can change or reset a forgotten password by email. The web portal shares one cache and one live poller between pages and sends 70% fewer requests. The driver app added password reset and booking prices from the server, and a YOLOv10 paper was reproduced for plate detection.

II. AI & Backend Progress

- **Plate reading service:** the plate detector and reader from last week now run in the AI service, so the backend can read plates from any camera image.
  - *Read plates:* takes an image and returns each plate's box, confidence, and the text in English and Bangla; weak plates are dropped.
  - *Logging:* every read is kept for 30 days for checking and later training.
- **Plate check:** the backend reads the plate of each occupied slot and compares it with that slot's booking.
  - *Live reads:* each slot stores the last plate text, confidence and time; an empty slot clears them.
  - *Result: match*, *mismatch* (a different car in a booked slot), *no booking*, or *unknown* when the plate cannot be read; it is returned with the owner's camera list.
- **Bookings module:** covers the whole booking from price to departure.
  - *Price quote:* shows the price before booking; a slot's own price comes first, then the vehicle rate, then the lot's base price. The app can no longer send a price itself.
  - *Booking list:* filter by status and date, with pages.
  - *Check-in:* by scanning the slot's QR code, or by phone location within 200 m of the lot.
  - *Extend and check-out:* extend while parked if the slot is free; a late check-out after 10 minutes of grace is charged per started hour.
  - *Cancel:* a paid booking is refunded.
  - *Automatic clean-up:* unpaid holds expire after 15 minutes, missed bookings become *no-show*, and cars not checked out an hour after the end are closed.
- **Payments module:** a test payment gateway, so the app and portal can build the full payment flow before a real gateway is connected.
  - *Start payment:* creates a payment for a booking and returns a test payment link.
  - *Confirmation:* the gateway's signed callback marks the booking paid; repeated callbacks are ignored.
  - *History:* a user can list their payments.
- **Accounts:** password and profile changes.
  - *Change password:* signs out the user's other devices.
  - *Forgot password:* a 6-digit code is emailed, valid 15 minutes with five tries.
  - *Profile:* checks phone numbers and role-specific fields; drivers can now open any approved lot with its prices and photos.
- **Database and testing:** two migrations for plate reads and the plate check, applied to the live database; old accounts and their data removed. 116 endpoint checks were run against the real database, all passing.

A. Plate Model Contribution: Method Proposal

The plate detector and reader are standard YOLO models trained on public data. To add a contribution of our own to the models themselves, we studied their errors, read 22 papers (six on Bangladeshi plates), and wrote the method proposal below.

- **Error study:** on 218 test plates the reader gets 75.7% of plates fully right with a 4.4% character error rate.
  - *Wrong length:* in the full test run, 93 of 212 wrong plates had the wrong number of characters: seven digits instead of six (the dash read as a one), a series letter found twice, or none at all.
  - *Dark plates:* a dim basement car park in Dhaka gave "Dhaka Metro 21-8842" with the letter missing.
  - *New sources:* a detector trained on one source only scored 0.991 mAP50 on its own test set but 0.910 on our merged one.
  - *Baseline check:* the vision-language reader of [5], reproduced on our data, read 31.2% of plates at 43.7 ms, so the YOLO reader (2.5 ms) stays the baseline.
- **Literature gap:** end-to-end readers, cascaded Bangla YOLO models [4], low-light enhancement, tilt correction [6] and layout rules [7] are all published. No Bangla work builds the plate layout into the model, handles low light or tilt inside it, removes train/test duplicates [8], or reports results across datasets.
- **Method 1, reading by redrawing:** BRTA fixes the plate design, so any plate number can be drawn in software.
  - *Fitting:* for each of the reader's top few readings, a small differentiable renderer draws that plate and fits its four corners, brightness and blur to the camera image with a few gradient steps; the reading whose drawing fits best is kept.
  - *Why:* tilt, low light and blur are handled by one fitting step; the leftover difference shows when a reading is

unreliable; the same renderer makes training plates for rare districts and letters.
  - *Novelty:* rendered plates have only been used as training data; we found no work that reads a plate by fitting a drawn plate to the image.
- **Method 2, vehicle-aware letter:** by public sources, the series letter depends on the vehicle (motorcycle A, Ha, La; private car Ka, Kha, Ga, Bha; CNG Tha; bus Ba), and the plate colour shows private or commercial use.
  - *Head:* a new detector head predicts vehicle type and plate colour, and the reader may only choose letters allowed for that vehicle.
  - *Many frames:* on traffic cameras the character scores of one tracked plate are combined over frames; the Bangla work we read uses single images.
- **Evaluation plan:** same data and splits, one change at a time: baseline, fixed plate rules, redrawing, vehicle head, both. Results reported for dark, tilted and small plates, across datasets, and on our own Dhaka camera, with plate accuracy, character error rate, mAP50 and time per plate.

III. Frontend Progress

- **Shared query cache:** one cache for the whole browser tab. Pages that need the same data share it instead of loading it again, requests for the same data at the same moment are sent once, and the last good data stays on screen when a reload fails.
- **One live poller:** the live occupancy map was polled separately by every portal page. Now all pages watch one poller that pauses while the tab is hidden, backs off after failures, and checks map points with no lots only every 5 minutes. In a 115 s test session over four pages it sent 9 requests instead of 30 for the same data, 70% fewer; a hidden tab now sends 0 requests instead of 10, and switching pages 0 instead of 5 (Fig. 1). The cache is about 310 lines of new code.
- **Tests and notes:** 9 unit tests for the cache, covering shared requests, stale data, failed reloads, polling start and stop, and sign-out. Before-and-after notes document the change.
- **Paper review:** SVTRv2 [1], a fast text reader from ICCV 2025, was reviewed as a plate reader. It resizes each crop by its shape, which suits two-row Bangla plates.

IV. Mobile Progress

- **Password:** a forgot-password screen sends the emailed code and sets a new password, and a change-password screen was added to the profile.
- **Booking price:** the booking sheet now asks the server for the price before booking, and booking details show subtotal and discount.
- **Requests:** the app's signed-in requests now also support sending data, with a retry after a token refresh.
- **Paper reproduction:** YOLOv10 [2], a detector that needs no box filtering step after prediction, was fine-tuned for plate detection and compared with YOLOv8 [3] under the same settings on 1,153 test images (Table I). Accuracy was the same; YOLOv10-N was fastest with TensorRT on a T4 and on CPU. YOLOv8-N found more small plates (60% recall against 20%) and read more plates exactly as a character reader (84.0% against 81.6%). Six models were trained, about 10 GPU hours in total.

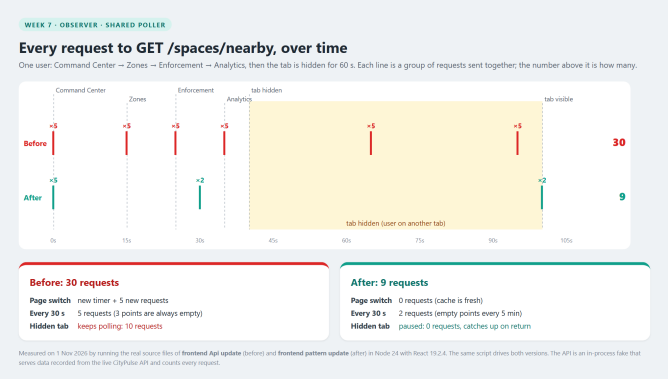


Fig. 1. Live occupancy requests in one portal session, before and after the shared poller.

TABLE I
PLATE DETECTION, YOLOV10 REPRODUCTION (1,153 TEST IMAGES)

Model	mAP50	Night	ms (T4, TRT)
YOLOv10-N	98.7%	96.7%	**6.00**
YOLOv10-S	98.2%	**96.9%**	7.08
YOLOv8-N	98.7%	94.5%	6.87
YOLOv8-S	98.5%	94.3%	8.19

V. Plan for Next Week

- Start the plate model contribution with the vehicle-type head, and test plate rules on the reader's output.
- Backend: notifications, a health check, camera video upload and delete, and jam-time offers.
- Frontend: owner camera and slot setup with plate check results; mobile: check-in, check-out and payment screens.

References

[1] Y. Du *et al.*, "SVTRv2: CTC beats encoder-decoder models in scene text recognition," in *Proc. ICCV*, 2025.
[2] A. Wang *et al.*, "YOLOv10: Real-time end-to-end object detection," in *Proc. NeurIPS*, 2024.
[3] G. Jocher, J. Qiu, and A. Chaurasia, "Ultralytics YOLO," 2023. [Online]. Available: https://github.com/ultralytics/ultralytics
[4] P. Sarker and S. M. S. Alam, "Cascaded YOLO-based approach for Bangladeshi license plate identification and number recognition," *PLOS ONE*, vol. 21, no. 8, 2026.
[5] N. Hasin *et al.*, "A robust deep learning framework for Bangla license plate recognition using YOLO and vision-language OCR," *arXiv:2603.10267*, 2026.
[6] S. M. Silva and C. R. Jung, "License plate detection and recognition in unconstrained scenarios," in *Proc. ECCV*, 2018.
[7] R. Laroca *et al.*, "An efficient and layout-independent automatic license plate recognition system based on the YOLO detector," *IET Intelligent Transport Systems*, vol. 15, no. 4, 2021.
[8] R. Laroca *et al.*, "Do we train on test data? The impact of near-duplicates on license plate recognition," in *Proc. IJCNN*, 2023.